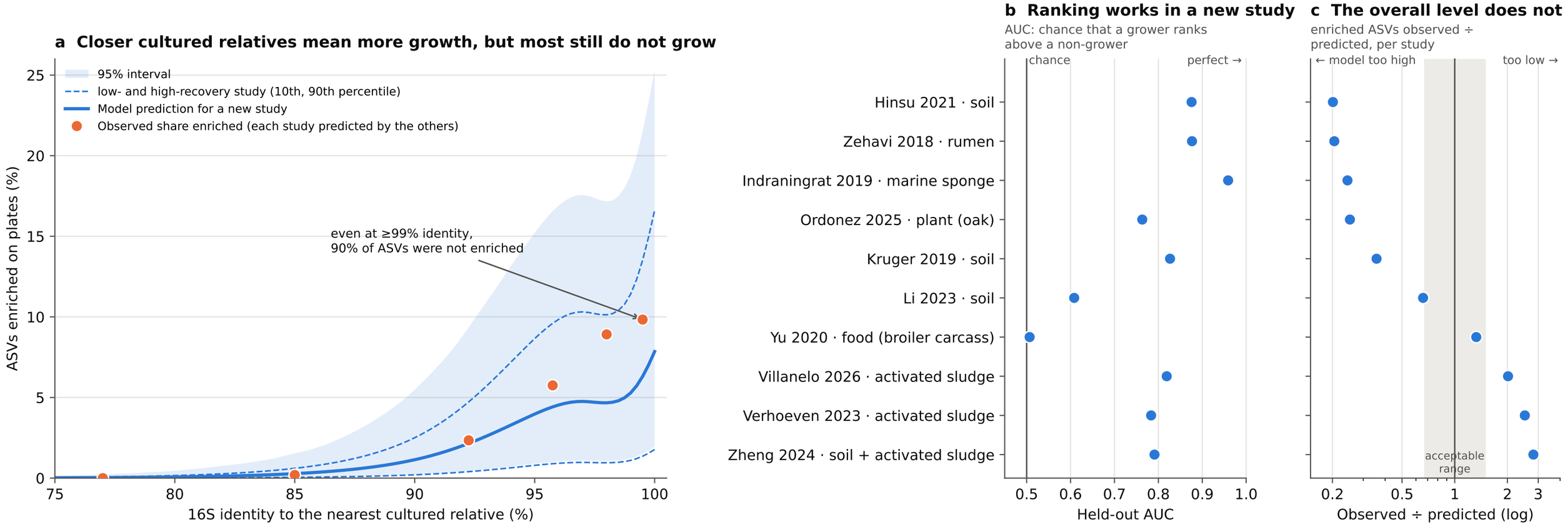
a Closer cultured relatives mean more growth, but most still do not grow
95% interval
low- and high-recovery study (10th, 90th percentile)
Model prediction for a new study
Observed share enriched (each study predicted by the others)
even at ≥99% identity,
90% of ASVs were not enriched
ASVs enriched on plates (%)
16S identity to the nearest cultured relative (%)
b Ranking works in a new study
AUC: chance that a grower ranks above a non-grower
chance
perfect →
Hinsu 2021 · soil
Zehavi 2018 · rumen
Indraningrat 2019 · marine sponge
Ordonez 2025 · plant (oak)
Kruger 2019 · soil
Li 2023 · soil
Yu 2020 · food (broiler carcass)
Villanelo 2026 · activated sludge
Verhoeven 2023 · activated sludge
Zheng 2024 · soil + activated sludge
Held-out AUC
c The overall level does not
enriched ASVs observed ÷ predicted, per study
← model too high
too low →
acceptable range
Observed ÷ predicted (log)